

Solvency Field Theory V12.13

Trace-Variational D-Scale Law and Conditional Top-Sector 6π Closure

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Generated 2026-06-19T03:24:33Z

Version status: Scoped freeze document. This document freezes a conditional D-scale trace law and a conditional top-sector 6π closure theorem candidate. It does not claim a complete particle mass hierarchy, species map, final top/e closure, residual-gap explanation, blind prediction suite, or complete global action theorem.

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Abstract

Solvency Field Theory V12.13 consolidates the V12.12 particle-program results into a scoped trace-variational framework. The document begins from the V12.12 locally real particle foundation and the V12.12A photon closure identity, where the photon wavelength at energy mc^2 equals the massive closed 4π spinor path. It then develops a local settlement-reinforcement picture in which particle persistence is not a race against cosmic expansion but a local balance between Compton-scale refresh and settlement relaxation. The D-scale law is frozen as a conditional trace-variational candidate,

$$M_i = G_i \exp \left[\left(3\pi - \frac{1}{2} \ln \frac{25}{13} \right) u_i \right],$$

where the coefficient is derived from protected 3+2 sector trace weighting rather than fitted to D. A conditional top-sector extension is then stated: at D-scale saturation, the least nonzero no-overwrite terminal completion that closes both real support sectors is one untaxed $K_{\text{top}} = 4\pi + 2\pi = 6\pi$ layer. This predicts a top/e-scale diagnostic within about 0.63% of the current context value. The result is a scoped framework, not a complete particle mass hierarchy: species assignment, residual closure, full global action proof, and prediction/null testing remain open.

1 Status and claim boundary

V12.13 is a scoped version document. It freezes a conditional D-scale trace law and a conditional top-sector 6π closure theorem candidate. It does not claim a complete particle mass hierarchy, species map, final top/e closure, residual-gap explanation, or blind prediction/null suite.

This version builds on the V12.12 locally real particle foundation: the particle is treated as real energy at c in real support motion, not as an abstract graph vertex with decorative fiber labels. The 4π term is real curvature crossing. The 2π term is real frame/gauge phase completion. The three spatial dimensions are real settlement geometry. Without that reality lock, the V12.13 quantities would be formal labels rather than physical support structure.

Claim boundary. V12.13 should be read as a scoped freeze of a candidate framework: trace-variational D-scale law plus conditional top-sector 6π closure. It is not a final particle mass hierarchy and it does not assign species.

2 Photon closure foundation from V12.12A

The closure-fit identity is

$$\lambda_\gamma(E = mc^2) = 4\pi r_{\text{spin}},$$

with

$$r_{\text{spin}} = \frac{\hbar}{2mc}.$$

Thus

$$4\pi r_{\text{spin}} = \frac{h}{mc} = \lambda_C.$$

This connects a photon open-path wavelength at energy mc^2 to the massive closed 4π spinor path. V12.13 uses this as the physical anchor for treating 4π as a real closure measure.

3 Local settlement reinforcement

R192M replaced the earlier Hubble-race intuition with a local maintenance equation: a particle persists by refreshing its local settlement/gravity support faster than local relaxation erases it. The important rate identity is

$$\frac{\Omega_r}{\nu_C} = 4\pi.$$

Interpretation: one Compton maintenance cycle samples a 4π curvature crossing. This makes 4π a local curvature-crossing rate, not a symbolic decoration.

4 Closure geometry: 4π , 2π , and 6π

The V12.13 closure spine uses two physically distinct closure components:

$$4\pi = \text{spatial/spinor curvature closure}, \quad 2\pi = \text{frame/gauge phase closure}.$$

Together,

$$\Delta_{\text{close}} = 4\pi + 2\pi = 6\pi.$$

The 4π term enters from local curvature crossing and photon/massive closure. The 2π term enters from frame/gauge phase holonomy. The full 6π combination is the natural support-frame terminal closure measure.

5 Directional settlement bridge

R195M supplies the directional-settlement bridge: the 2π phase sector is included as a settlement load because frame/gauge orientation is not merely bookkeeping. Directional settlement compatibility gives it mechanical content: same circulation orientation produces a repulsive compatibility pattern, while opposite orientation meshes as an attractive compatibility pattern. In V12.13 language, charge/frame orientation creates real directional settlement structure that participates alongside scalar depth settlement. This is why the phase/frame sector contributes a real trace weight rather than remaining an abstract fiber label.

This bridge is not a full Coulomb derivation or a complete force law. Its role here is narrower: it justifies treating the 2π frame/gauge completion as physical support structure in the trace-variational D-scale law.

6 The $3 + 2$ canonical sector split

R208M-R210M supply the $3 : 2$ origin chain. The D-scale law depends on a protected $3 + 2$ split:

$$3 \text{ spatial settlement directions}, \quad 2 \text{ phase/frame closure directions}.$$

The effective participation number is

$$N_{\text{eff}} = \frac{25}{13} = 1.923076923076923.$$

The amplitude tax is

$$\frac{1}{2} \ln \left(\frac{25}{13} \right) = 0.326963233703332.$$

The effective load is

$$L_{\text{eff}} = 3\pi - \frac{1}{2} \ln \left(\frac{25}{13} \right) = 9.097814727066048.$$

The important feature is that these numbers come from internal sector counting and trace weighting. They are not fitted to D.

7 D-scale trace-variational law

The frozen D-scale candidate is

$$M_i = G_i \exp(L_{\text{eff}} u_i),$$

with

$$L_{\text{eff}} = 3\pi - \frac{1}{2} \ln \left(\frac{25}{13} \right).$$

At the endpoint used in the D diagnostic,

$$D_{\text{scale}} = 17871.488623573841.$$

The comparison is

$$D_{\text{target}} = 17877.208689571427, \quad \frac{D_{\text{scale}}}{D_{\text{target}}} = 0.999680035843575,$$

with percent error

$$-0.031996415642\%.$$

This is a diagnostic agreement, not a definition.

Independence and non-fitting boundary. The central V12.13 coefficient is not defined by D, top/e, the R172 selected family, or any measured particle mass. The coefficient

$$L_{\text{eff}} = 3\pi - \frac{1}{2} \ln \left(\frac{25}{13} \right)$$

is constructed from internal structure only: 3π from the half-action/closure-load lane, and $25/13$ from protected $3+2$ sector trace weighting. The comparison to D is diagnostic, not definitional. The exact top/e gap is likewise not used to define the top-sector law.

8 R211M-R217M proof chain

The D-scale law is supported by the following chain:

1. R211M: action-level 3 : 2 grouping proof attempt.
2. R212M: separable projector trace route.
3. R213M: normalization and cross-term audit.
4. R214M: trace-protection mechanism via symmetry/incommensurability.
5. R215M: canonical sector normalization.
6. R216M: master-action canonical $3+2$ patch.
7. R217M: second-variation trace-weight derivation.

Together, these runs support the conditional D-scale coefficient $L_{\text{eff}} = 3\pi - \frac{1}{2} \ln(25/13)$. R218M freezes this as a D-scale law candidate, not a final mass hierarchy.

9 Conditional top-sector 6π closure

R219M-R223M extend the D-scale framework to a conditional top-sector completion. The top-sector theorem candidate is:

Given: A1. D-scale trace lane is saturated. A2. No-overwrite minimality applies. A3. Terminal completion must close both real support sectors. A4. Full closure measure is $4\pi + 2\pi$. A5. Terminal completion is untaxed. Then the unique minimal nonzero terminal completion is $K_{\text{top}} = 4\pi + 2\pi = 6\pi$.

R223M tightens the premises: A1 receives an action-boundary route using a normalized u domain with $0 \leq u \leq 1$ and saturation at $u = 1$; A3 is promoted to a formal action-route candidate because terminal closure cannot leave either real support sector open; A5 is promoted to an action-category separation candidate because branch participation tax and terminal completion are different action categories.

The candidate top-sector output is

$$D_{\text{scale}} \times 6\pi = 336869.624211198883.$$

Compared to the top/e context,

$$\text{top/e context} = 339000.0, \quad \text{factor} = 0.9937157056377755,$$

with percent error

$$-0.628429436225\%.$$

The residual is

$$\text{exact gap} - 6\pi = 0.119205279071775, \quad \text{exact gap percent above } 6\pi = 0.632403646897\%.$$

The residual remains open and is not fitted.

10 Combined candidate law

The scoped V12.13 candidate can be summarized as

$$M_i = G_i \exp(L_{\text{eff}} u_i) K_{\text{top}}^{I_{\text{top}}},$$

where

$$L_{\text{eff}} = 3\pi - \frac{1}{2} \ln \left(\frac{25}{13} \right), \quad K_{\text{top}} = 4\pi + 2\pi = 6\pi.$$

Here I_{top} is an activation flag, not a fitted continuous exponent. It means the top-sector terminal completion is applied only when the D-scale trace lane is saturated and the top-sector closure condition is active.

11 What is not claimed

V12.13 does not claim:

- a complete particle mass hierarchy,
- a derived species map,
- final top/e closure,
- residual-gap explanation,
- blind prediction/null suite,
- complete global action theorem,
- measured-mass fit.

The D-scale and top/e comparisons are diagnostic only.

12 Open problems

The main open problems are:

1. derive A1 saturation from the global action;
2. derive A3 terminal necessity from the global action;
3. derive A5 action-category separation explicitly;
4. explain or demote the 0.6324% residual above 6π ;
5. build the species-map constraint skeleton;
6. build the blind prediction/null suite;
7. decide whether V12.13 remains a scoped candidate framework or becomes the foundation for a later species program.

13 Preservation statement

SFT V12.13 freezes a scoped trace-variational D-scale law and conditional top-sector 6π closure framework. The D-scale coefficient $L_{\text{eff}} = 3\pi - \frac{1}{2} \ln(25/13)$ is derived from protected $3+2$ sector trace weighting, not from D or measured masses. The top-sector extension states that, at D-scale saturation, the least nonzero no-overwrite terminal completion closing both real support sectors is one untaxed $K_{\text{top}} = 4\pi + 2\pi = 6\pi$ layer. The resulting D-scale endpoint agrees with the D diagnostic to about 0.032%, and the conditional top-sector 6π output agrees with the top/e context to about 0.63%. These agreements are diagnostic, not definitional. Species assignment, residual closure, prediction/null testing, and full global action proof remain open.

A Equation index

ID	Name	Equation	Source
E1	Photon closure identity	$\lambda_\gamma(E = mc^2) = 4\pi r_{\text{spin}}$	V12.12A
E2	Spin radius	$r_{\text{spin}} = \frac{\hbar}{2mc}$	V12.12A/R192M
E3	Curvature crossing rate	$\Omega_r/\nu_C = 4\pi$	R192M
E4	Full closure load	$\Delta_{\text{close}} = 4\pi + 2\pi = 6\pi$	R192M- R194M/R220M
E5	Effective feedback load	$L_{\text{eff}} = 3\pi - \frac{1}{2} \ln\left(\frac{25}{13}\right)$	R208M-R217M
E6	D-scale law	$M_i = G_i \exp(L_{\text{eff}} u_i)$	R211M-R218M
E7	Top-sector theorem candidate	$K_{\text{top}} = 4\pi + 2\pi = 6\pi$	R221M-R223M
E8	Combined candidate	$M_i = G_i \exp(L_{\text{eff}} u_i) K_{\text{top}}^{I_{\text{top}}}$	V12.13 synthesis

B Run catalog

V12.12 / R184R-R185F

Reality lock foundation; Main. Locally real particle foundation: particles are real energy at c in real support motion, not abstract graph vertices.

V12.12A

Photon closure addendum; Main. Photon open-path wavelength at energy mc^2 equals massive closed 4π spinor path.

R187T / V12.12B	<i>Compression scaffold; Setup.</i> Toy compression scaffold; no species map promoted.
R189M	<i>D-span clarification; Setup.</i> D is base dynamic range, not full top/e span.
R190M-R191M	<i>K/beta and half-action clues; Brief clue.</i> K_req, beta, sqrt2, and half-action patterns identified as clues, not final law.
R192M	<i>Local settlement reinforcement; Main.</i> Particle maintenance as local reinforcement vs relaxation; curvature crossing rate gives 4pi.
R193M	<i>2pi frame/gauge holonomy; Main.</i> 2pi phase/gauge closure component.
R194M	<i>Support-fiber closure bridge; Main.</i> 4pi+2pi=6pi support/frame closure candidate.
R195M	<i>Directional settlement bridge; Main paragraph.</i> Frame/gauge orientation contributes real directional settlement structure; 2pi enters as physical settlement load.
R208M-R210M	<i>Two-channel / 3:2 origin; Main bridge.</i> 3 spatial : 2 phase grouping leads to N_eff=25/13.
R211M-R217M	<i>D-scale proof chain; Core.</i> Action grouping, projector trace, normalization, trace protection, canonical patch, and second variation.
R218M	<i>D-scale freeze; Main.</i> D-scale trace law frozen as conditional candidate.
R219M-R223M	<i>Top-sector 6pi conditional extension; Main.</i> Gap triage, formal 6pi, activation criterion, symbolic theorem, and premise closure audit.
R188M/R198M/R200M-R204M	<i>Exploratory routes; Appendix/omit.</i> Simple exponent, Green-solver, inverse/feedback explorations redirected the route but are not needed in main derivation.
R196M-R197M	<i>Future force/charge lanes; Future work.</i> Coulomb/null and relaxation-kernel lanes remain useful for later force program, not core V12.13.

C Claim boundary matrix

Allowed	V12.13 gives a scoped trace-variational D-scale law and conditional top-sector 6pi closure framework.
Allowed	The D-scale coefficient is derived independently of D and top/e.
Allowed	The top-sector 6pi layer is a symbolic conditional theorem candidate.
Blocked	The exact top/e gap defines the law.
Blocked	The complete particle mass hierarchy is derived.
Blocked	A species map is derived.
Blocked	The top/e scale is fully closed.
Blocked	The 0.6324 percent residual above 6pi is explained.
Blocked	A blind prediction/null suite is complete.
Blocked	The global master-action theorem is complete.

D Open claims ledger

A1 global saturation theorem

Open. D-scale saturation as u=1 endpoint has an action-boundary

route, but global variational derivation remains open.

A3 terminal necessity

Open. Both-sector terminal closure is a formal action route; final theorem remains open.

A5 tax split

Open. Terminal completion $N_{\text{eff}}=1$ and $\text{tax}=0$ has action-category support; explicit action split remains open.

Residual gap

Open. Exact gap is 0.6324 percent above 6π ; not explained or fitted.

Species map

Open. No particle species assignments are derived.

Prediction/null suite

Open. No blind no-retuning suite yet.

Full hierarchy closure

Open. No complete particle mass hierarchy claim.